

The University of Azad Jammu and Kashmir, Muzaffarabad.



Computer Networks

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Star Topology Using Switch

Introduction

A star topology is a network configuration in which all devices (computers, printers, etc.) are connected to a central device called a switch. The switch manages communication between devices, making the network efficient, reliable, and easy to troubleshoot.

To design and configure a star topology using a switch and assign IP addresses to ensure communication between devices.

Procedure

● Step 1: Open Simulation Software

Open Cisco Packet Tracer and prepare a blank workspace for network design.

● Step 2: Add Devices (Star Layout)

Place one switch at the center and add multiple PCs around it to form a star topology.

● Step 3: Connect Devices to Switch

Connect each PC to the switch using straight-through cables (all links should turn green).

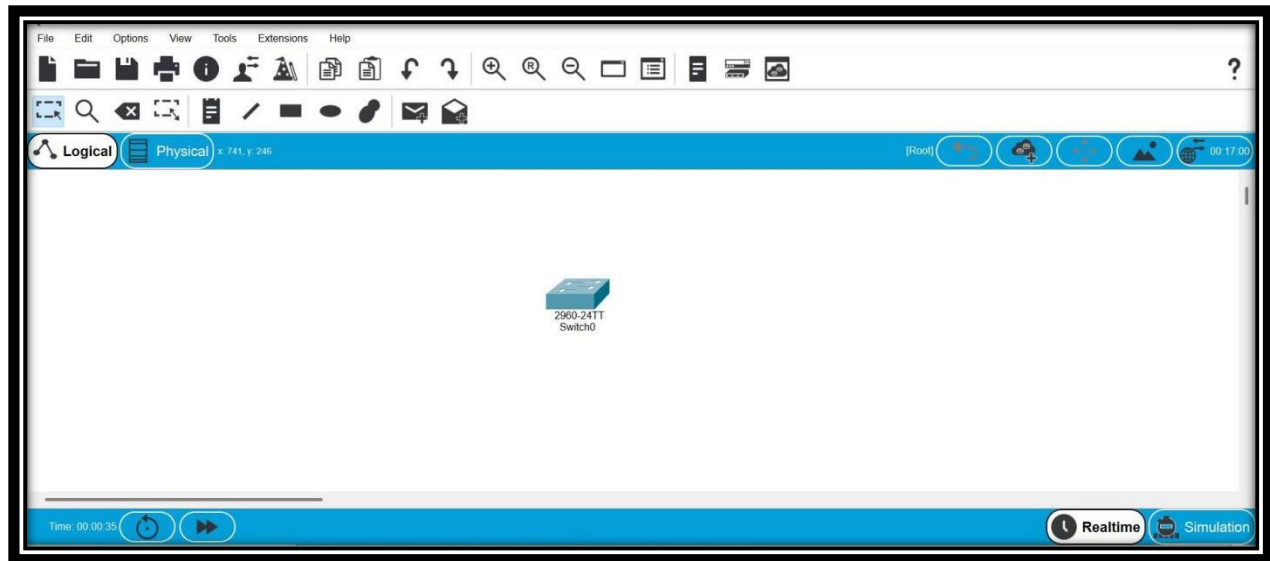
● Step 4: Configure IP Addresses

Assign unique IP addresses to each PC within the same network (e.g., 192.168.1.x with subnet mask 255.255.255.0).

● Step 5: Verify Connectivity Using Ping and Packet Simulation

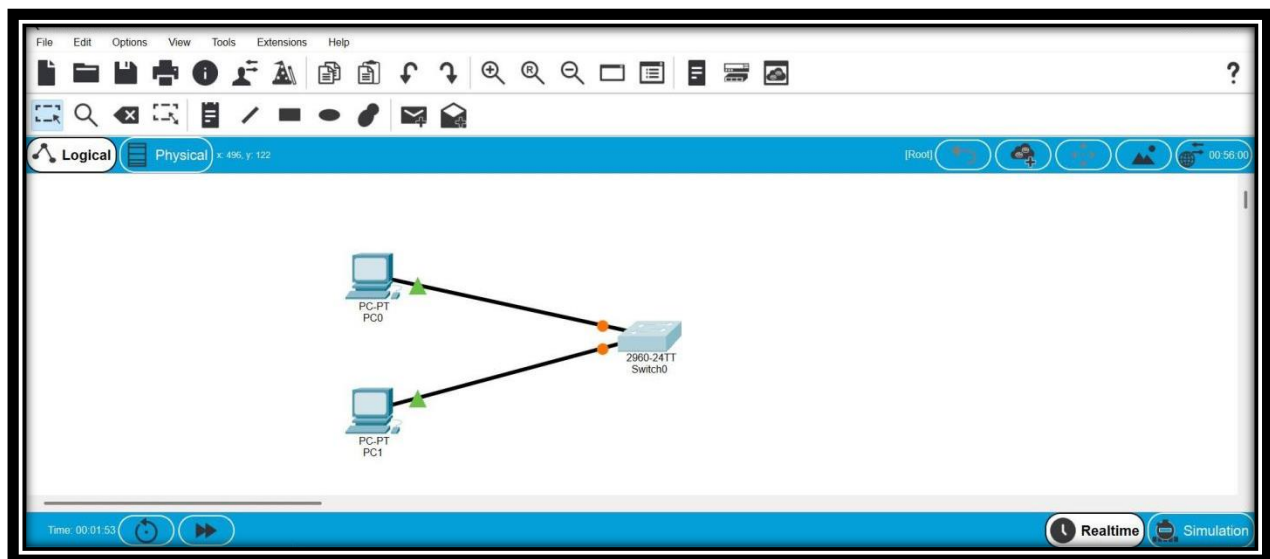
Use the ping command between PCs and also test using simulation mode to observe packet flow through the switch.

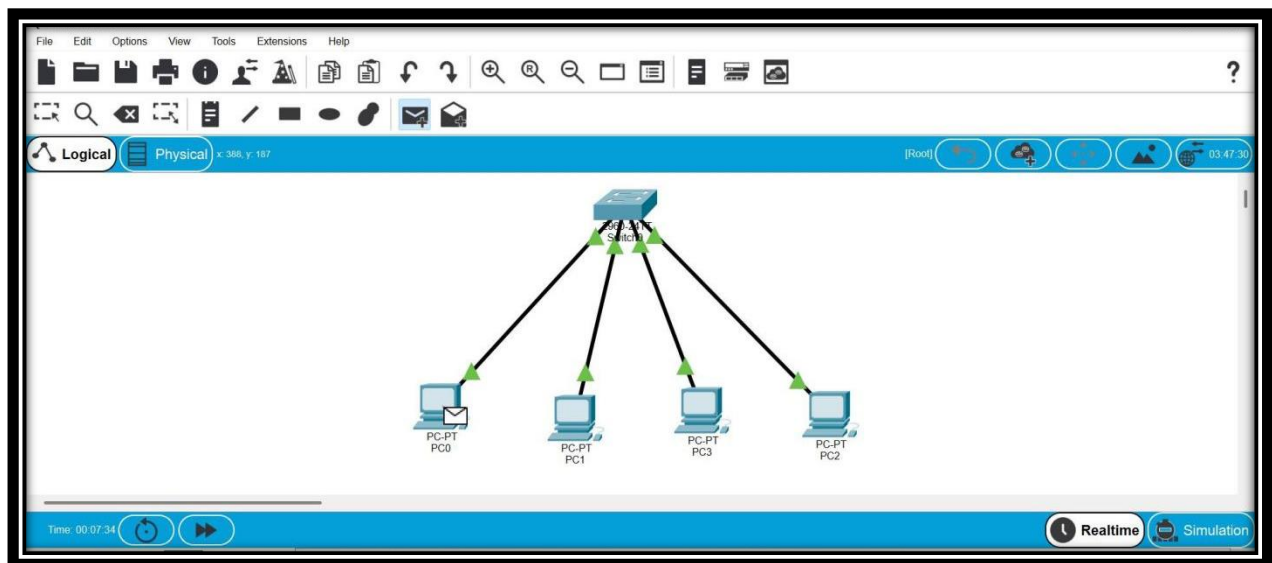
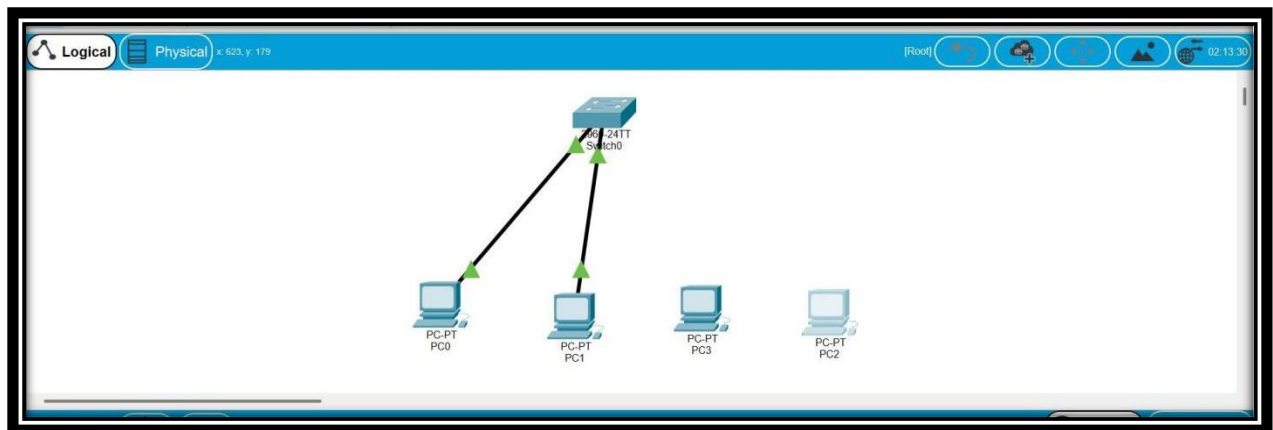
Step 1 & 2: Adding Switch and PCs (Star Layout)



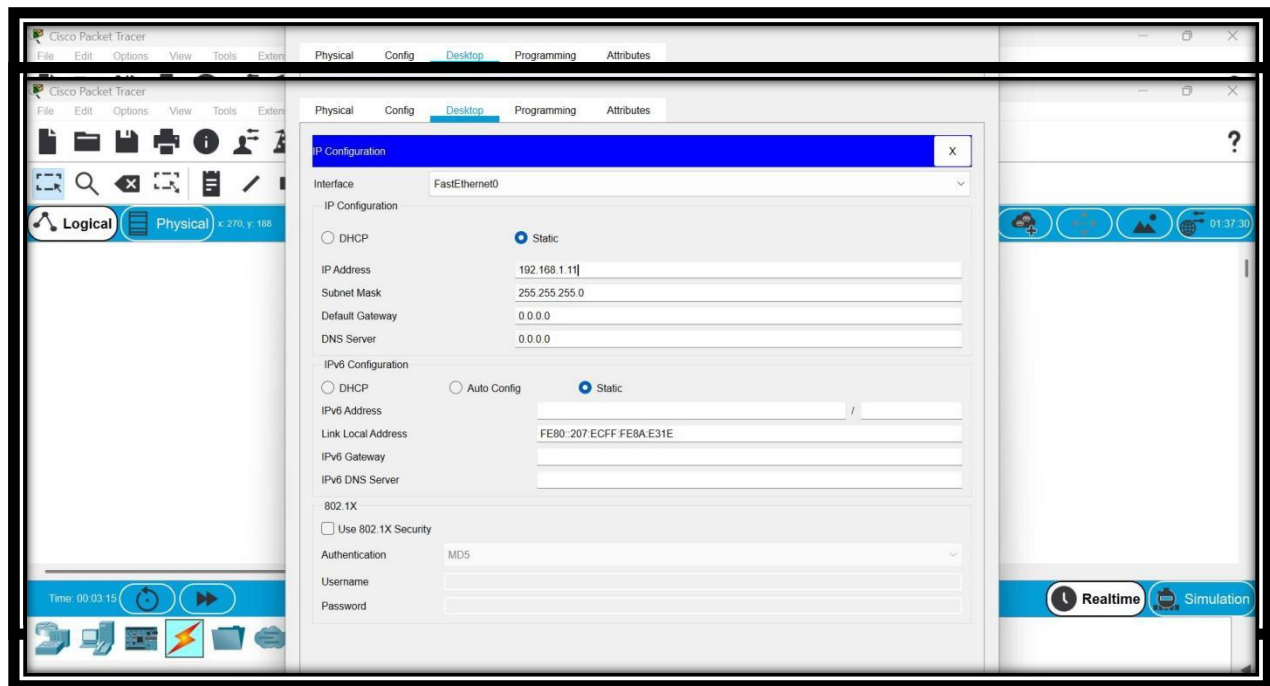


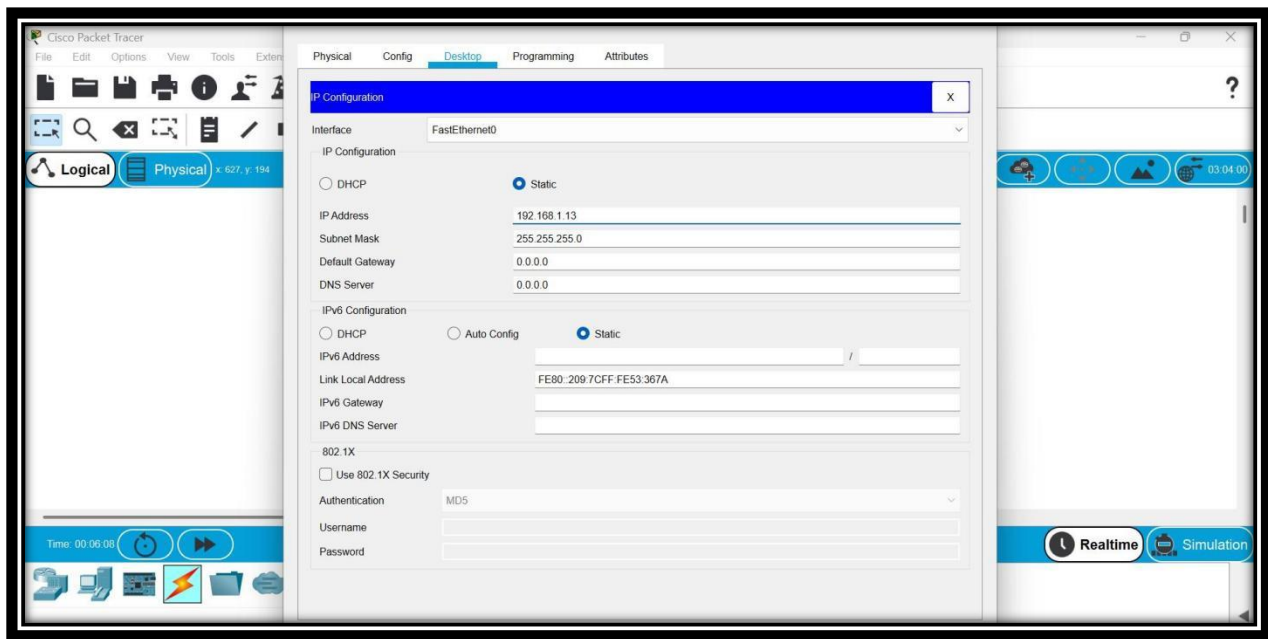
Step 3: Connecting Devices to Switch





● Step 4: Configuring IP Addresses (Multiple PCs)





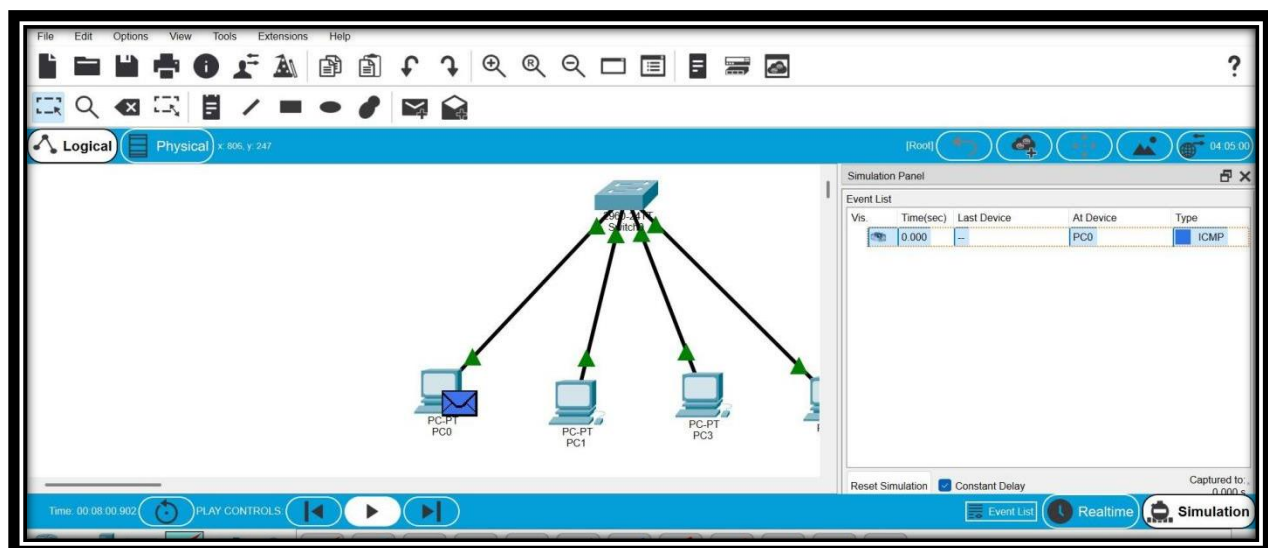
- **Step 5: Ping Results and Packet Simulation**

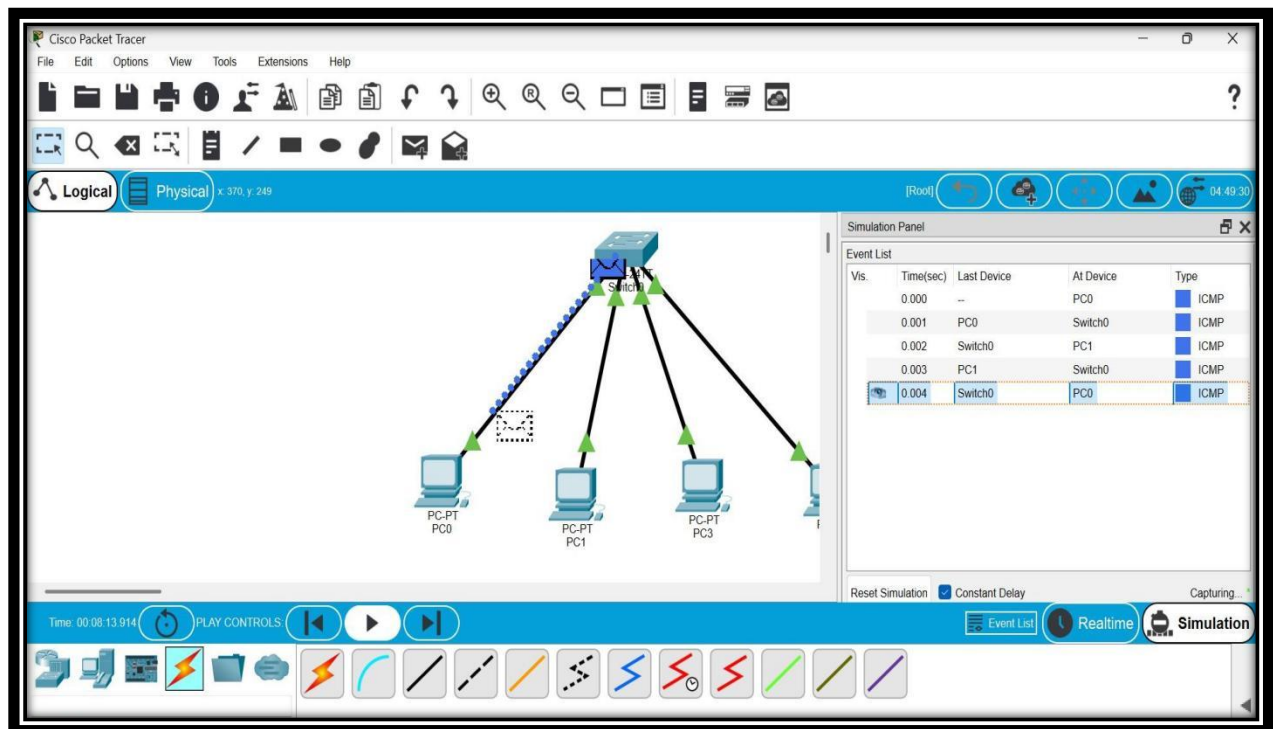
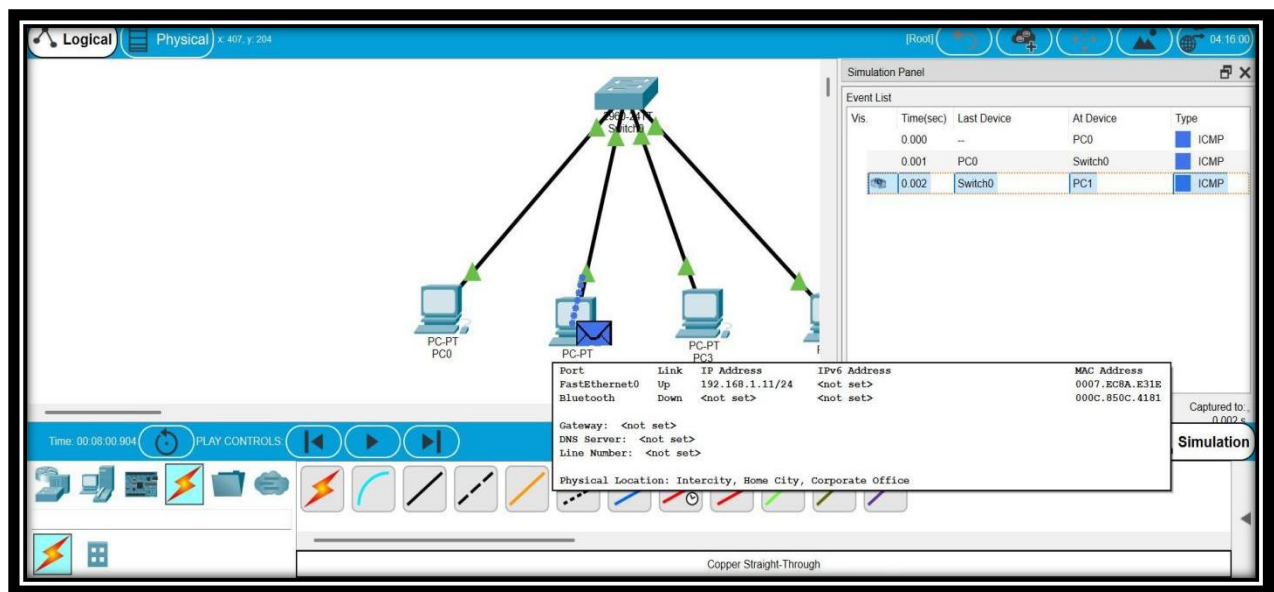
```
C:\>ping 192.168.1.2

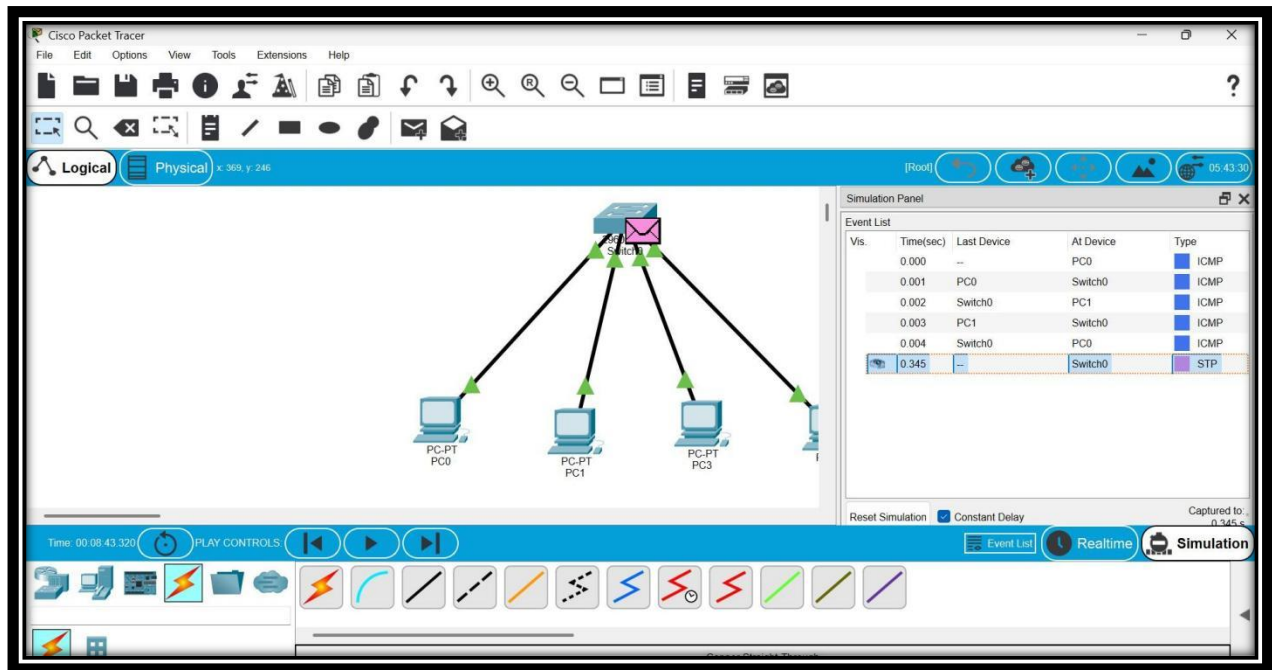
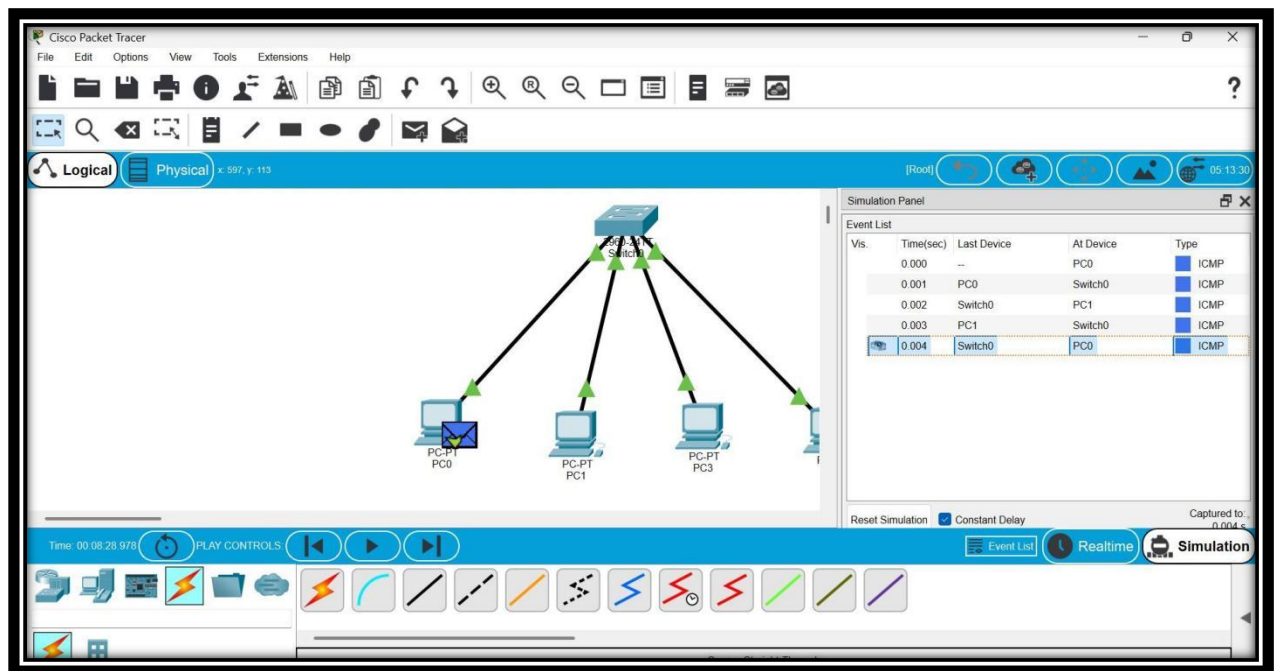
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```







Conclusion

The star topology using a switch provides efficient communication and centralized control. It is easy to manage and troubleshoot, making it one of the most commonly used network topologies.